Table of Contents

Software Requirements Specification (SRS)

Project: ResiTrack — Apartment / Residencies Organizer **Format:** IEEE 830-1998 **Course:** Software Engineering

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1. Introduction

1.1 Purpose

Specifies the software requirements for **ResiTrack**, a web-based apartment and residency organizer, for use by the development team and course evaluators.

1.2 Scope

Apartment complexes currently manage tenant registration, rent tracking, maintenance, complaints, visitor logs, and notices through disconnected manual processes, causing communication gaps and poor visibility. ResiTrack (built on the MERN stack) centralizes these operations into one platform for four actors — **Administrator, Resident, Maintenance Staff, Owner**.

1.3 Definitions, Acronyms, Abbreviations

Term	Definition
FR / NFR	Functional / Non-Functional Requirement
RBAC	Role-Based Access Control
UC	Use Case
MERN	MongoDB, Express.js, React, Node.js
TLS	Transport Layer Security
PII	Personally Identifiable Information

1.4 References

- IEEE Std 830-1998
- `TestPlan.docx`, `ArchitectureDesign.docx` (companion deliverables)

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2. Overall Description

2.1 Product Perspective

A new, browser-accessible 3-tier system (React client / Express API / MongoDB) replacing manual and informal (phone, paper) coordination between residents and management.

2.2 Actors

#	Actor	Responsibilities
1	**Administrator**	Tenant registration, rent tracking, notices, flat management, Kanban oversight, documents, receipts.
2	**Resident (Tenant)**	Login, maintenance requests, complaints, visitor logging, notices, rent receipts.
3	**Maintenance Staff**	Views assigned requests; updates status (In Progress / Resolved).
4	**Owner**	Read-only: financial summaries, occupancy reports.
5	**System** (automated)	Payment reminders, RBAC enforcement.

2.3 Operating Environment

Evergreen browsers (desktop + mobile, ≥375px width) · Node.js/Express server on free-tier cloud hosting · MongoDB Atlas · HTTPS/TLS 1.2+ throughout.

2.4 Constraints & Assumptions

MERN stack and MVC architecture are mandatory; no licensed third-party software; single semester (~16 weeks) delivery; single-complex deployment (no multi-tenancy); users have basic digital literacy.

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3. Specific Requirements

3.1 Functional Requirements

Each requirement is written to be clear, unambiguous, and independently verifiable.

ID	Requirement	Priority	Verification Criterion
FR-01	Admin shall register	High	Record appears in DB

ID	Requirement	Priority	Verification Criterion
	a new tenant with flat number, contact details, and lease start/end dates.		with all fields populated.
FR-02	Tenants shall log in using registered email and password.	High	Valid credentials succeed; invalid credentials are rejected.
FR-03	Admin shall post community notices visible to all residents.	High	Notice visible to all logged-in residents within 5s of posting.
FR-04	Residents shall raise maintenance requests with category, description, and priority.	High	Request saved with status “Open”; confirmation shown.
FR-05	Maintenance staff shall update request status (Open / In Progress / Resolved).	High	Status change reflected in DB and resident view within 3s.
FR-06	Admin shall record and track monthly rent payments per tenant.	High	Payment record stores amount, date, flat ID; tenant marked paid.
FR-07	System shall send automated rent reminders for upcoming/overdue payments.	Medium	Reminder triggered ≥3 days before due date.
FR-08	Residents shall log visitor entries (name, contact, purpose, flat number).	Medium	Entry appears in admin dashboard with timestamp.
FR-09	Admin shall view a Kanban-style dashboard of open maintenance requests.	Medium	Board loads; cards draggable across status columns.
FR-10	Owners shall view financial summaries (rent collected, dues, occupancy rate).	Medium	Dashboard totals match DB aggregation.

ID	Requirement	Priority	Verification Criterion
FR-11	System shall support document upload for lease agreements and ID proofs.	Medium	PDF/JPEG ≤5 MB accepted, stored, retrievable.
FR-12	Admin shall manage flat inventory (add / edit / deactivate units).	High	Change reflected immediately in tenant assignment.
FR-13	Tenants shall file complaints and track resolution status.	Medium	Complaint logged; status transitions visible to resident.
FR-14	System shall enforce role-based access control (Admin, Resident, Maintenance, Owner).	High	Out-of-role request returns HTTP 403.
FR-15	System shall generate printable monthly rent receipts (PDF).	Low	PDF contains tenant name, flat, amount, date.

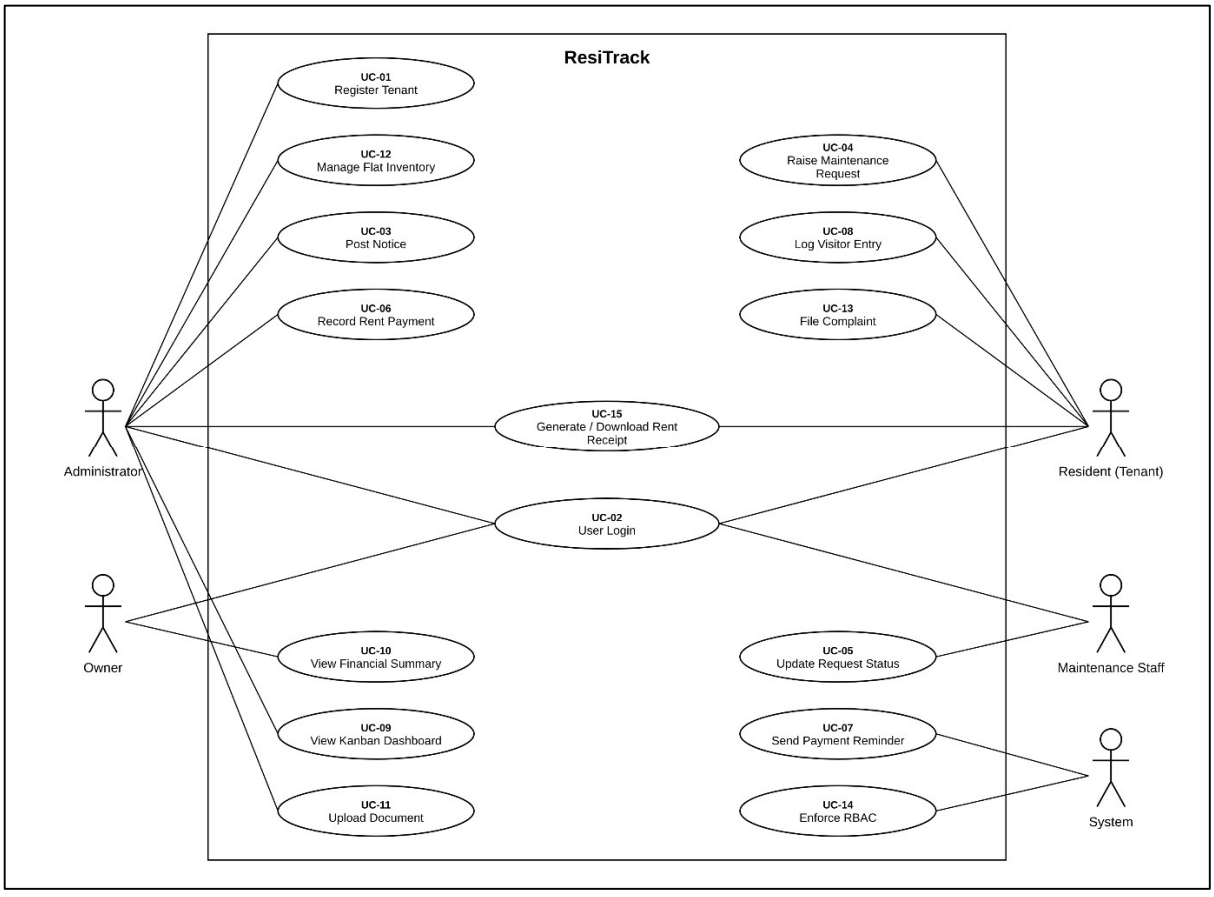
3.2 Non-Functional Requirements

ID	Requirement	Category	Verification Criterion
NFR-01	Respond to any user action within 3s under normal load (≤200 concurrent users).	Performance	Load test: 95th-percentile ≤3s at 200 vUsers.
NFR-02	99.5% availability during peak hours (8 AM–10 PM).	Availability	30-day uptime monitor ≥99.5%.
NFR-03	Passwords stored as bcrypt hashes (cost factor ≥12).	Security	DB shows hash prefix `$2b$12$`, never plaintext.
NFR-04	All data in transit encrypted (TLS 1.2+).	Security	SSL scan grade A; plaintext HTTP rejected.
NFR-05	Session times out after 30 minutes of inactivity.	Security	Idle session token rejected; redirect to login.
NFR-06	UI usable on mobile browsers	Usability	Renders correctly at 375px viewport.

ID	Requirement	Category	Verification Criterion
	(responsive).		
NFR-07	New admin completes onboarding + adds a tenant in ≤10 minutes unaided.	Usability	Timed usability test.
NFR-08	Scales to 500 tenant records without response-time degradation.	Scalability	Perf test at 500 records: ≤3s response.
NFR-09	Codebase follows MVC; ≥60% unit test coverage.	Maintainability	Coverage report + architecture review.
NFR-10	All admin actions (login, edit, delete) are logged for audit.	Auditability	Log entry created and retrievable per action.

4. Use Case Diagram

Shared use cases — **UC-02 (Login)**, **UC-14 (RBAC)**, **UC-15 (Rent Receipt)** — serve multiple actors.



ID	Requirement	Category	Verification Criterion

Use Case Diagram

UC ID	Use Case	Actor(s)
UC-01	Register Tenant	Administrator
UC-02	User Login	All
UC-03	Post Notice	Administrator
UC-04	Raise Maintenance Request	Resident
UC-05	Update Request Status	Maintenance Staff
UC-06	Record Rent Payment	Administrator
UC-07	Send Payment Reminder	System
UC-08	Log Visitor Entry	Resident
UC-09	View Kanban Dashboard	Administrator
UC-10	View Financial Summary	Owner
UC-11	Upload Document	Administrator

UC ID	Use Case	Actor(s)
UC-12	Manage Flat Inventory	Administrator
UC-13	File Complaint	Resident
UC-14	Enforce RBAC	System
UC-15	Generate/Download Rent Receipt	Administrator, Resident

5. Security Section

5.1 Security Objectives

#	Objective
SO-01	**Confidentiality** — tenant PII (contact details, lease/ID documents) accessible only to authorized roles, never exposed in transit or at rest.
SO-02	**Integrity** — payment records, maintenance status, and audit logs modifiable only through authorized, attributable actions.
SO-03	**Availability of role-appropriate access** — each actor reliably accesses only its permitted functions, with predictable denial otherwise.
SO-04	**Accountability** — every privileged action traceable to an authenticated user and timestamp.

5.2 Security Requirements

ID	Requirement	Objective
SEC-01	Passwords stored as bcrypt hashes, cost factor ≥12 (= NFR-03).	SO-01, SO-04
SEC-02	All data in transit encrypted via TLS 1.2+ (= NFR-04).	SO-01
SEC-03	RBAC middleware returns HTTP 403 for any out-of-role request (= FR-14).	SO-03
SEC-04	Sessions expire after 30 minutes of inactivity (= NFR-05).	SO-01, SO-03
SEC-05	Admin actions logged immutably with actor ID and timestamp (= NFR-10).	SO-02, SO-04

ID	Requirement	Objective
SEC-06	Uploaded documents restricted to PDF/JPEG, ≤5 MB (= FR-11).	SO-01, SO-03

6. Requirements Traceability Matrix (RTM)

Req ID	Use Case	Test Case ID
FR-01	UC-01	TC-01
FR-02	UC-02	TC-02
FR-03	UC-03	TC-03
FR-04	UC-04	TC-04
FR-05	UC-05	TC-05
FR-06	UC-06	TC-06
FR-07	UC-07	TC-07
FR-08	UC-08	TC-08
FR-09	UC-09	TC-09
FR-10	UC-10	TC-10
FR-11	UC-11	TC-11
FR-12	UC-12	TC-12
FR-13	UC-13	TC-13
FR-14	UC-14	TC-14
FR-15	UC-15	TC-15
NFR-01	—	TC-NFR-01
NFR-02	—	TC-NFR-02
NFR-03	UC-02	TC-NFR-03
NFR-04	—	TC-NFR-04
NFR-05	UC-02	TC-NFR-05
NFR-06	All UCs	TC-NFR-06
NFR-07	UC-01, UC-02	TC-NFR-07
NFR-08	—	TC-NFR-08
NFR-09	—	TC-NFR-09
NFR-10	All Admin UCs	TC-NFR-10